

Resource Allocation for Metaverse Experience Optimization: A Multi-Objective Multi-Agent Evolutionary Reinforcement Learning Approach

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Abstract—In the Metaverse, real-time, concurrent services such as virtual classrooms and immersive gaming require local graphic rendering to maintain low latency. However, the limited processing power and battery capacity of user devices make it challenging to balance Quality of Experience (QoE) and terminal energy consumption. In this paper, we investigate a multi-objective optimization problem (MOP) regarding power control and rendering capacity allocation by formulating it as a multi-objective optimization problem. This problem aims to minimize energy consumption while maximizing Meta-Immersion (MI), a metric that integrates objective network performance with subjective user perception. To solve this problem, we propose a Multi-Objective Multi-Agent Evolutionary Reinforcement Learning with User-Object-Attention (M2ERL-UOA) algorithm. The algorithm employs a prediction-driven evolutionary learning mechanism for multi-agents, coupled with optimized rendering capacity decisions for virtual objects. The algorithm can yield a superior Pareto front that attains the Nash equilibrium. Simulation results demonstrate that the proposed algorithm can generate Pareto fronts, effectively adapts to dynamic user preferences, and significantly reduces decision-making time compared to several benchmarks.

Index Terms—Metaverse experience, multi-objective optimization, resource allocation, Meta-Immersion, energy consumption.

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I. INTRODUCTION

THE Metaverse is characterized by its immersive nature, allowing seamless integration between physical and virtual experiences, offering users the ability to live, work, and play in a digital environment [1]. For example, hands-on training with immersive simulations enables workers to refine their skills within a risk-free virtual realm [2]. Using virtual classrooms would allow students and teachers to interact in 3D virtual spaces for lectures, discussions, and collaborative activities [3]. Compared to the passive experience of live video streaming, the Metaverse is distinguished by its high level of interactivity. Real-time immersive Metaverse services place rigorous requirements on hardware and software to provide low delay and fast rendering performance. In terms of hardware, lightweight, low-power head-mounted displays (HMDs) and physical assistive devices are required to use high-resolution images for a long time without frequent charging or battery replacement [4]. In terms of software, rising delay adversely affects the user experience, potentially leading to dizziness and motion sickness due to sensory confusion [5]. Due to the limitations in computing power and battery capacity of HMD devices, balancing terminal energy consumption while ensuring immersive user experience and achieving high-quality, low-latency rendering on low-power devices has become an urgent challenge to address.

Meta-experience refers to the overall quality and depth of a user's experience when interacting within virtual environments. For the Metaverse to provide immersive and continuous services, user preferences for the meta-experience and energy demand can vary depending on the distinct scenarios and objectives of the Metaverse users (MUs) involved. In scenarios focused on immersion, users prioritize a rich and engaging meta-experience, even if it consumes more energy. On the other hand, when battery life is limited or in power-constrained environments, saving energy becomes more important than the meta-experience. In this regard, dynamic user preferences, random mobility, and changing channel conditions require adaptability. This ensures users have high-quality and satisfying meta-experience. At the same time, it promotes sustainable usage, even in varying access situations.

The experience in virtual space and the terminal energy consumption are greatly affected by the power allocated by the base stations (BSs). Additionally, rendering capacity decisions

play a significant role. Rendering capacity refers to the system's ability to process and generate visual content in real-time. These decisions impact the computing power of the MU hardware device. For typical Metaverse applications where MUs participate in virtual spaces under resource constraint wireless environment [6], [7], the power allocation and rendering capacity activation decision among multiple MUs is essentially a cooperative game. However, due to the limited resources and simultaneous requests from multiple MUs, achieving this optimal balance and stability of resource allocation poses several challenges: 1) How to achieve fairness and stability in resource allocation among multiple users? Fairness indicates that the resource allocation decision does not favour specific users. Stability refers to the balanced state of resource allocation among users with their preferences. Each MU can maximize individual objects while taking into account the behaviour and influence of other MUs. 2) How to solve the multi-objective optimization problem (MOP) of balancing meta-experience and terminal energy consumption for individual MUs? To enable ubiquitous and embodied access that requires extensive computational resources, leveraging ubiquitous distributed computing power can mitigate straggler effects as it relieves base stations' computing workload. Given the Metaverse user cases stated before, distributed computing becomes a natural choice, which also prompts us to develop an algorithm that rationally allocates transmission power and makes rendering capacity activation decisions in response to dynamic changes in user preferences and environmental uncertainty.

To tackle the aforementioned challenges, we propose a novel algorithm that exploits an evolutionary algorithm combined with multi-agent reinforcement learning (MARL) to optimize resource allocation under dynamic environment with diverse user preferences. The key contributions of this paper can be outlined as follows:

- We establish an MOP and yield Pareto-optimal solutions that model and represent trade-offs between meta-experience and terminal energy consumption in wireless multi-user networks. By taking into account the downlink rate, uplink bit error probability and subjective feelings of MUs, we design a Meta-Immersion (MI) metric to evaluate meta-experience via Weber–Fechner Law.
- To solve the intractable MOP, we propose the multi-objective multi-agent evolutionary reinforcement learning with user-object-attention (M2ERL-UOA) algorithm with decentralized execution to allocate the transmission power and rendering capacity. We also introduce the novel user-object-attention mechanism to reduce the large action space of the problem. By training discrete policy networks for Pareto front points rather than a single large model for the entire Pareto region, our approach narrows the search space and better reduces resource consumption. To our knowledge, this is the first work that applies M2ERL to the Metaverse over wireless communications.
- Performance of M2ERL-UOA is evaluated through rigorous theoretical analysis. Theoretical studies demonstrate that, in the case of reduced action space and state space, the policy network of each point on the Pareto front generated

by the proposed algorithm can optimal value of the objective function and reach the Nash equilibrium in wireless multi-user networks.

- We use the prediction model to find the high-quality Pareto front and exploit the adaptability and generalization capabilities of neural networks to address resource allocation under different user preferences. Simulation results show that our algorithm can shorten the policy decision delays satisfying the strict requirement of Metaverse services compared with the traditional evolutionary algorithms and other baseline optimization algorithms. Specifically, our scheme outperforms other methods in terms of Pareto performances between MI and consumption.

The remainder of the paper is structured as follows. In Section II, we provide a summary of existing research. Section III outlines the system model and wireless transmission model and formulates the optimization problem. In Section IV, we introduced the proposed M2ERL-UOA algorithm in detail. Additionally, Section V presents and discusses the simulation results. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Quality of Experience (QoE) in Metaverse

As the concept of user experience in Metaverse applications evolves, measuring QoE has become increasingly challenging due to its subjective nature. Traditional methods primarily focus on objective network parameters such as delay [8], downlink rate [15], [28], and uplink packet error rate (PER) [29], [30]. Unlike live video streaming that emphasizes continuous playback and visual clarity, the Metaverse requires higher downlink rate and lower delay to enable smooth and immersive interactions with altering virtual objects and environments. While live video streaming primarily involves passive data reception, the high interactivity of the Metaverse requires maintaining a low uplink PER to ensure accurate and responsive interactions between users and the virtual environment [31]. Higher human-centric utility corresponds to better QoE, contrasting with Quality of Service (QoS), which quantifies the system's objective performance. The authors in [18] assessed the QoE based on the mean squared error between an initial field of view (FoV) and the distorted FoV. Some studies also investigated methods for measuring human-centric subjective user perception (SUP). For instance, Zhao et al. [16] formulated human-centric utility as a logarithmic function of transmission rate and resolution. Implicit indicators, such as physiological, cognitive, and emotional behaviors, offers a less biased assessment of QoE. Using sensors, wearable devices, and eye-tracking tools, users' physiological parameters and cognitive states can be captured in real-time and non-invasively [32], [33].

B. Resource Allocation in the Metaverse Over Wireless Communications

Resource allocation in Metaverse networks presents a multifaceted challenge, requiring careful balancing of multiple objectives to ensure optimal performance. Recent advancements,

including multi-dimensional resource optimization strategy presented by [8], have focused on enhancing the meta-experience of MUs engaged in wireless mixed reality services, with QoE serving as the primary optimization criterion. Similarly, Liu et al. [12] addressed resource management in wireless virtual reality (VR) networks enhanced by edge computing, reinforcing the importance of QoE in these strategies. Despite the significant advancements in works such as [8], [11], [12], these studies often consider QoE as a standalone objective, overlooking the intricate interactions between QoE and other critical factors, including latency, energy consumption, and cost.

In real-world Metaverse deployments, devices and resources are inherently constrained, necessitating a more holistic approach to resource allocation. For example, the Stackelberg game-theoretic policy-based learning algorithm proposed in [15] have shown excellent anti-jamming performance in resource allocation. Zhao et al. [16] considered human-centric QoE and system costs, optimizing the utility-cost ratio (UCR) in wireless VR networks using a novel fractional programming technique. Si et al. [17] introduced a communication resource allocation algorithm based on outer approximation (OA) to achieve an optimal balance between latency and QoE. Zhan et al. [18] proposed an energy-efficient reinforcement learning based resource allocation scheme to fulfill the QoE against jamming. Zhang et al. [24] efficiently met users' pre-defined QoE requirements for video quality and latency tolerance through rational offloading decisions and timely cache updates. The user-centric critic with heterogeneous actors algorithm in [27] allocates transmission channels and power to meet frame rate and resolution requirements while reducing transmission failures. Nonetheless, for real-time interactive Metaverse applications that necessitate both optimal user experiences and high energy efficiency, existing research remains limited.

C. Multi-Objective Optimization in Metaverse Networks

The dynamic nature of Metaverse systems being Markovian across continuous time slots presents a partially observable Markov decision process (POMDP) problem, which is typically difficult to solve. Traditional optimization methods, including convex optimization [10], [17], heuristics [25], and game theory [13], [21], may require global information to achieve an optimal solution, which not only increases communication overhead but also adds computational complexity. Unlike traditional methods, multi-objective reinforcement learning (MORL) can handle complex control problems using prior information extracted from dynamic Metaverse scenarios. This is because reinforcement learning (RL) methods can quickly adapt their behaviour to changing environments through interaction with the corresponding environment. However, several works like [14], [18], [20], [23], [26], [27], [34] all employ single-policy methods, where each run uses a scalarized function to convert a multi-objective problem into a single-objective one, and then solve it with conventional RL algorithms. This approach fails to address conflicts among objectives, as MUs' preferences for different objectives are typically unknown in advance. The weighted sum approach is typically biased, making it difficult to achieve a

balance between objectives. In contrast, multi-policy methods (e.g., [19], [24]) can learn a set of policies approximating the true Pareto front, where each policy corresponds to a different trade-off, allowing decision-makers to choose a policy that matches their current preferences. However, these algorithms address multi-objective problems (MOPs) for single agents. When facing the diverse preferences of multiple users under resource constraints, single-agent multi-policy MORL algorithms typically struggle to achieve Nash equilibrium among users. The differences of this paper from prior works are summarized in Table I.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we introduce the system model for the meta-experience with the closed-form expression of MI metric [9]. We then formulate the multi-objective optimization problem concerning the trade-off of MI and energy consumption. The main notations and descriptions are enumerated in Table II.

In this paper, we consider a scenario where a wireless network provides the communication resources to MUs, denoted by $\mathcal{K} = \{1, 2, \dots, K\}$, for virtual immersive experience. MUs in the system have different preferences for meta-experience and terminal energy consumption at different times. Fig. 1 depicts the system model. The transmit power allocated by the BS and rendering capacity to the MUs determine the rendering effect of virtual objects and the real-time performance of system interaction. The MUs have a variety of options for virtual immersion experiences, such as virtual travel, virtual reality games, and virtual fitness. These experience scenes are composed of individual virtual objects, which are equivalent to a material library, and the terminal uses this material library to build a complete experience scene. In this paper, N_k^O is used to represent the quantity of virtual objects for the k th user.

A. Wireless Transmission Model

Ensuring the dependable transmission of user-object interactions and achieving low-latency delivery of virtual object data are crucial for improving the meta-experience. Therefore, this paper adopts ultra-reliable low-latency communications (URLLC) [35] technology to meet the needs. Under URLLC scenario, the transmission rate can be expressed follows the finite code length transmission model [36], which establishes the mathematical relationship between transmission rate, code length, and PER. For MU k , with the code length b and downlink error probability E_k^D , the downlink transmission rate R_k^D is given by

$$R_k^D \approx \log_2(1 + \zeta) - \sqrt{\frac{1}{b} \left(1 - \frac{1}{(1 + \zeta)^2}\right)} \frac{Q^{-1}(E_k^D)}{\ln 2}, \quad (1)$$

where ζ donates signal-to-noise ratio and $Q^{-1}(\cdot)$ is the inverse function of $Q(\cdot)$, which can be expressed as $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}} dt$. With the code length b and the uplink transmission rate R_k^U , the PER of the entire uplink transmission E_k^U

TABLE I
EXISTING WORKS OF RESOURCES ALLOCATION FOR THE METAVERSE

Ref.	Optimization Problem			Solutions	Dynamic Impact Factors		Action Space Dimension Reduction	Equilibrium Among Users	Application
	Optimization Objectives	Multi-Objective Problem	Perferences		User Preferences	Channel Quality			
[8]	QoE (Delay)	×	-	MADRL	×	✓	×	✓	MR
[9]	QoE (BEP, Rate, SUP)	×	-	Heuristic Algorithm	×	✓	✓	×	Metaverse
[10]	QoE (Video quality, Stall time)	✓	linear, fixed	Convex Optimization	×	✓	×	×	VR
[11]	QoE (Delay, Socity distance, Video quality)	×	-	Game Theory	×	✓	×	✓	Metaverse
[12]	QoE (Video quality, Interaction latency)	×	-	DRL	×	✓	×	×	VR
[13]	Qos (Delay, Tracking accuracy)	✓	linear, dynamic	Game Theory, ESN	✓	✓	✓	✓	VR
[14]	Qos (Delay, PER, Throughput)	✓	linear, fixed	MARL	×	✓	×	✓	VR
[15]	Cost, QoE (Rate)	×	-	MARL	×	✓	×	✓	Metaverse
[16]	Cost, QoE (Delay, Video quality)	×	-	Fractional Programming	×	✓	×	×	Metaverse
[17]	QoE (Rate), Energy	✓	linear, fixed	Convex Optimization	×	✓	×	×	MAR
[18]	QoE (FoV, Delay), Energy	✓	linear, fixed	DRL	×	✓	×	×	Metaverse
[19]	Delay, Cost	✓	(Non)linear	MORL	✓	✓	×	×	Metaverse
[20]	Delay, Energy	✓	linear, dynamic	MARL	×	✓	×	×	Metaverse
[21]	Delay, Energy	✓	linear, dynamic	Game Theory, RL	×	✓	✓	✓	Metaverse
[22]	Delay, Cost	✓	(Non)linear	Evolutionary Algorithm	✓	✓	×	×	VR
[23]	Delay, Energy	✓	linear, dynamic	DRL	×	✓	✓	×	VR
[24]	Delay, QoE (Video quality)	✓	(Non)linear	MORL	✓	✓	×	×	VR
[25]	Energy, VR content distortion	✓	linear, dynamic	Heuristic Algorithm	×	✓	×	×	VR
[26]	Delay, Mean accuracy, Mean absolute error	✓	linear, fixed	MARL	×	×	×	×	Metaverse
[27]	Energy, QoE (Rate, Delay)	✓	linear, fixed	MARL	×	✓	×	✓	Metaverse
Our Work	Energy, QoE (BEP, Rate, SUP)	✓	(Non)linear	M2ERL-UOA	✓	✓	✓	✓	Metaverse

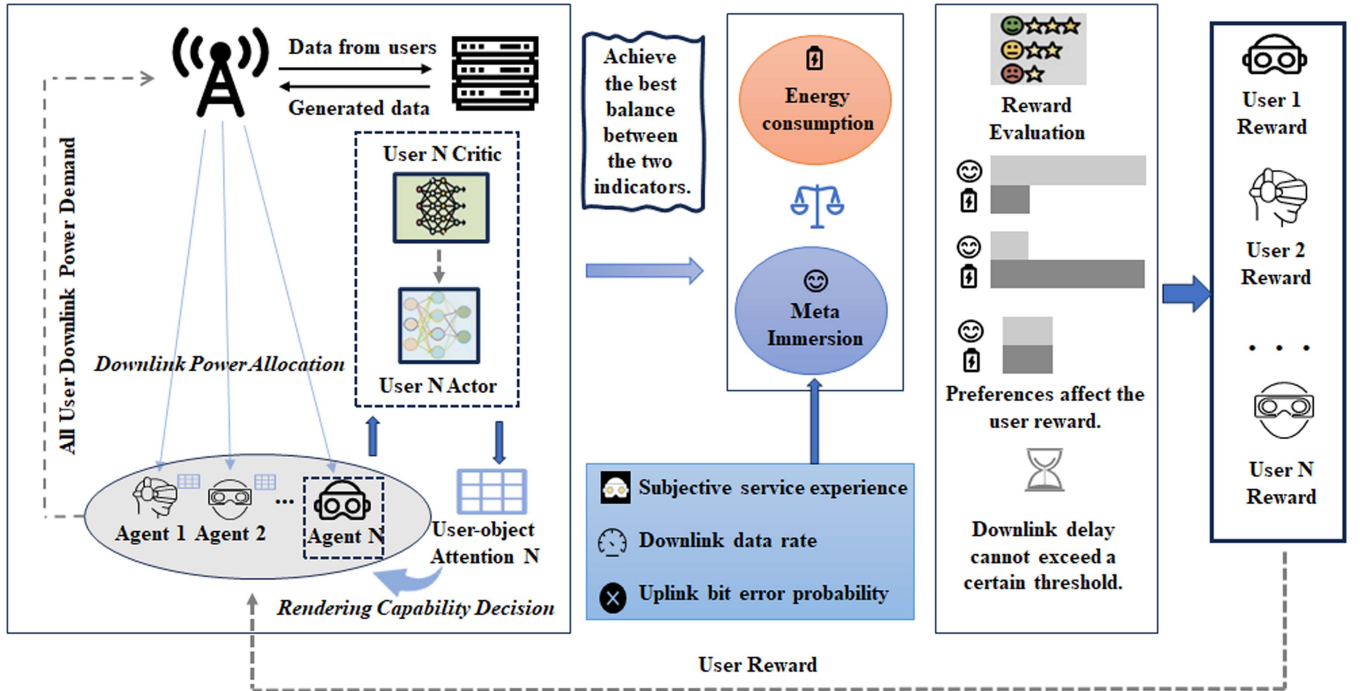


Fig. 1. System model. This figure illustrates how the BS allocates downlink power while MUs make attention-aware rendering capability and power allocation decisions. The system evaluates multiple indicators based on user preferences, achieving a balance between energy consumption and Meta-Immersion.

TABLE II
SUMMARY OF MAIN NOTATIONS AND DESCRIPTIONS

Notation	Description
Notation used in system model	
$\mathcal{K}$	The set of all MUs
N_k^O	The number of virtual objects for the k th user
R_k^D, R_k^U	The downlink/uplink transmission rate for the k th user
E_k^D, E_k^U	The downlink/uplink error probability for the k th user
P_{total}^D, P_{max}^U	The total downlink power and the maximum uplink power
P_k^D, P_k^U	The downlink/uplink power for the k th user
W_k^{DU}	The total energy consumption for the k th user
W_{max}	The maximum energy consumption
W_k	The energy consumption reward for the k th user
$T_{k,n}$	The k th user's attention to the n th object
$C_{k,n}^R$	The rendering capacity assigned to the n th virtual object by the k th user
C_{th}^R	The minimal rendering capacity threshold
C_k^R	The total render capacity allocated to the k th user
C_{max}^R	The maximum total rendering capacity of each user
M_k	The MI of the k th user
D_{max}	The maximum delay threshold
μ_k	The allocation policy adopted by the k th user
$\mathcal{N}_k$	The interference noise power of the k th user
$\mathcal{V}$	The number of bits per pixel
$\mathcal{B}$	Bandwidth of a Metaverse user terminal
Notation used in reinforcement learning	
a_k	Action of the k th user
a^t	The joint action space of all MUs at time step t
s_k	State of the k th user
s^t	The joint state space of all MUs at time step t
r_k^t	The reward of the k th user at time step t
μ_k^{θ}	The policy network of the k th user
V_k^{ϕ}	The critic network of the k th user
m_w	The number of RL iterations in the warm-up stage
m_t	The number of RL iterations in each generation during the evolutionary stage
ς	The weight of the sparsity metric
L	The number of learning tasks and elite offspring
γ	Discount factor

is calculated as follows:

$$E_k^U \approx \mathcal{Q} \left(\frac{\log_2(1 + \varsigma) - R_k^U}{\log_2 e \sqrt{\frac{1}{b} \left(1 - \frac{1}{(1+\varsigma)^2} \right)}} \right). \quad (2)$$

The total terminal energy consumption includes downlink energy consumption and uplink energy consumption. When the experience time of user k is t_k , the total terminal energy consumption is calculated as follows:

$$W_k^{DU} = (P_k^D + P_k^U) t_k. \quad (3)$$

where P_k^D and P_k^U represent the downlink power and the uplink power for the user k , respectively.

B. User-Object-Attention Mechanism

The MU's eye movements serve as a dependable reflection of attention allocation. Generally, the longer people stare at an object, the higher the level of attention to that object. A head-mounted displays worn by the MU facilitates the gathering of eye movement data [37]. An abundant dataset of eye movement proves valuable for analyzing the k th user's attention to the n th object which is denoted as $T_{k,n}$ ($k \in \{1, \dots, K\}, n \in \{1, \dots, N_k^O\}$). Rendering resources can be decided according to the attention of each object. Let $C_{k,n}^R$ denote the rendering capacity assigned to the n th virtual object by the k th user. More rendering resources are decided to objects with high user attention, and fewer rendering resources are decided to objects with low user attention. Through this approach, the enhancement of the meta-experience is achieved.

C. Meta-Immersion Metric

MI refers to the comprehensive feeling that MUs get during the whole virtual experience process [9]. Within the Metaverse service, MUs' experience hinges on three pivotal emotions: the feeling of presence, interaction and pleasure [38]. To precisely gauge these feelings and evaluate the corresponding QoS, we analyze both network performance and subjective experiences to calculate the MI of MU k .

We employ the downlink data rate denoted as R_k^D and the uplink bit error probability represented by E_k^U as metrics for assessing the network performance. The high downlink data rate facilitates low-latency data transmission, while the low uplink bit error probability ensures accurate user-object interactions. These two factors serve as key objective performance indicators within URLLC, exerting a pivotal influence on the interaction between virtual and reality.

The subjective service experience serves as a measure of the perceived quality within the Metaverse service. In the context of the Metaverse service explored in this paper, the subjective service experience is determined by the rendering quality, denoted as $C_{k,n}^R$. The Weber-Fechner Law [39] is employed to quantify the subjective feelings of MUs, stating that the perceived intensity or strength of a sensation is linearly related to the logarithm of the magnitude of the corresponding physical stimulus. The MI of the k th user is given as follows [9]:

$$M_k = \underbrace{\mathcal{L}(R_k^D) \mathcal{L}(1 - E_k^U)}_{\text{network performance}} \underbrace{\sum_{n=1}^{N_k^O} T_{k,n} \ln \left(\frac{C_{k,n}^R}{C_{th}^R} \right)}_{\text{subjective feeling}}, \quad (4)$$

where C_{th}^R represents the minimal rendering capacity threshold. The function $\mathcal{L}(\cdot)$ is employed to mitigate the influence of magnitudes [40], which is defined as

$$\mathcal{L} = \frac{l - l_{\min}}{l_{\max} - l_{\min}}, \quad (5)$$

where $l_{\min}$ and $l_{\max}$ indicate the minimum threshold and maximum value that the BS can provide, respectively. The first two terms of MI represent observable network performance, and the last characterizes subjective feeling. To ensure consistency in

magnitude with the first two terms, we use $T_{k,n}$ to quantify the subjective feeling. By this means, the multiplication operation combining the terms for calculating MI becomes more justifiable.

D. Problem Formulation

When MUs enter Metaverse for interactive experience, both MI and terminal energy consumption are critical factors affecting user satisfaction and sustainability. This necessitates considering a dynamic trade-off between these two metrics, as MUs' preferences vary with the specific usage scenario and their device's real-time battery status. Therefore, we initially formulate the optimization problem concerning power allocation and rendering capacity decisions as a MOP problem, which can be articulated as follows:

$$\begin{aligned} \max \quad & F(\boldsymbol{\mu}) = [f_1(\boldsymbol{\mu}), f_2(\boldsymbol{\mu})] \\ \text{subject to} \quad & \boldsymbol{\mu} \in \Omega, \end{aligned} \quad (6)$$

where $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_k]$ and $F(\boldsymbol{\mu})$ represent a k -dimensional decision vector and 2-dimensional objective function, respectively. $\Omega \subseteq \mathbb{R}^k$ and $F: \Omega \subseteq \mathbb{R}^2$ represents the corresponding attainable set in the objective space $\mathbb{R}^2$.

The first objective is to maximize MI described in (4),

$$\mathcal{O}_1 : \max \quad f_1(\boldsymbol{\mu}) = \sum_{k=1}^K M_k. \quad (7)$$

Since high energy consumption causes the device battery to drain quickly, it limits the duration of the MUs' virtual experience. In this paper, the second object is to minimize energy consumption formulated as follows:

$$\mathcal{O}_2 : \max \quad f_2(\boldsymbol{\mu}) = \sum_{k=1}^K W_{\max} - W_k^{DU}, \quad (8)$$

where $W_{\max}$ indicates the maximum energy consumption when the BS provides the maximum uplink power and the maximum downlink power to the terminal.

The k -th MU adopts the allocation policy $\mu_k \in \boldsymbol{\mu}$ to obtain the corresponding power and rendering capacity, and then the MI M_k and terminal energy consumption W_k can be calculated. We decomposing the MOP problem into individual optimization at each MU, enabling a more tailored optimization process that considers the unique conditions and preferences of each MU. The objectives and constraints at the user level are similar in nature but customized to account for individual user settings and system dynamics. Therefore, we propose a general optimization problem for each MU as follows:

$$\begin{aligned} \max \quad & F(\mu_k) = [M_k, W_k]. \\ \text{s.t.} \quad & C1. \quad \sum_{k=1}^K P_k^D \leq P_{total}^D, \\ & C2. \quad P_k^U \leq P_{\max}^U, \quad \forall k \in \mathcal{K}, \end{aligned}$$

$$\begin{aligned} C3. \quad & \sum_{n=1}^{N_k^O} C_{k,n}^R \leq C_{\max}^R, \quad \forall k \in \mathcal{K}, \\ C4. \quad & \frac{\mathcal{V} \sum_{n=1}^{N_k^O} C_{k,n}^R}{R_k^D} \leq D_{\max}, \quad \forall k \in \mathcal{K}. \end{aligned} \quad (9)$$

where C1 indicates that all downlink power allocated by the BS to all MUs cannot exceed its own total downlink power P_{total}^D . Constraints C2 and C3 indicate that the uplink power and total rendering capacity of each user cannot exceed their respective maximum values $P_{\max}^U$ and $C_{\max}^R$, respectively. We utilize rendering capacity, measured in pixels, and downlink rate to calculate latency. The stronger the rendering capacity of the terminal, the more data is transmitted, and the greater the delay is when the downlink rate remains the same. Constraint C4 indicates that the delay of each user cannot exceed a maximum delay threshold $D_{\max}$, ensuring a low delay of the user experience. $\mathcal{V}$ is the number of bits per pixel used to convert the rendering capacity to bits and figure out the delay.

The devices' charging status during operation introduces uncertainty, which could lead MUs to adjust their preferences for the two objectives. In the case that the MU is charging while experiencing, energy consumption is not the main consideration, but improving the MU's experience as much as possible. During data transmission, the interference noise power will also change due to changes in temperature, electromagnetic interference and other factors. Therefore, the real problem to be solved in this paper is how to flexibly decide uplink power and rendering capacity, and allocate downlink power for each MU dealing with the dynamic system preference and interference noise power, to maximize the obtain reward of the whole system.

IV. MULTI-OBJECTIVE MULTI-AGENT EVOLUTIONARY REINFORCEMENT LEARNING ALGORITHM WITH USER-OBJECT-ATTENTION

In this section, we propose the M2ERL-UOA algorithm to generate the uplink power and rendering capacity decision, and the downlink power allocation under dynamic interference noise power and user preference. Fig. 2 shows the general flow of the proposed algorithm.

A. Formulating MOP Into POMDP Model

In the scenario of this paper, the MUs do not have access to global information and can only perceive the environment through their local observations. A MARL algorithm is designed to solve the problem efficiently and accurately, where each MU acts as an agent. The Metaverse device decides uplink power and total rendering capacity based on the actions generated from its local observations. The actions are then sent to the BS, which allocates downlink power to each user. This multi-agent task is generally termed as partially observable Markov decision process (POMDP), which is described by $\{\mathcal{S}, \mathcal{A}, \mathcal{O}, r, \mathbb{P}\}$. Here, $\mathcal{S}$ represents the global state space of the environment and $\mathcal{O} = \{s_1, \dots, s_i, \dots, s_K\}$ is the partial observation of MUs. $\mathcal{A} = \{a_1, \dots, a_i, \dots, a_K\}$ is the joint action space of all MUs,

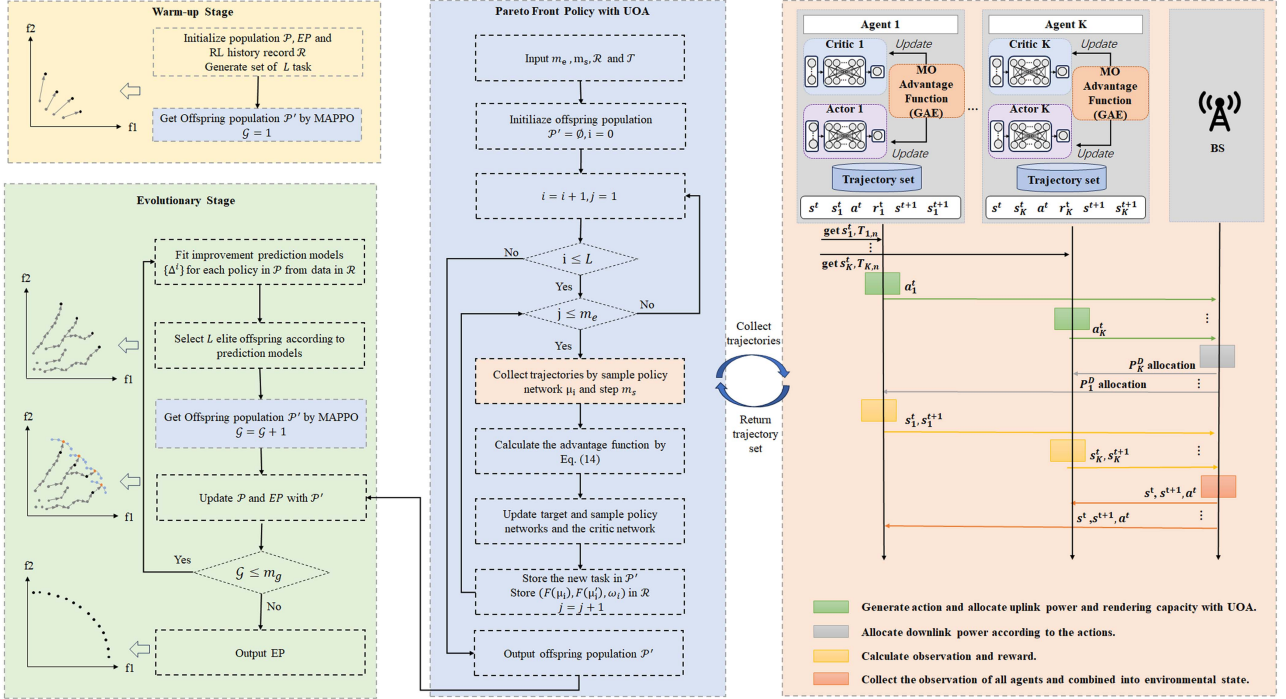


Fig. 2. Overview of the M2ERL-UOA algorithm.

where a_i denotes the individual action of the i th MU. One-step reward is expressed by r . $\mathbb{P}$ signifies the set of possibilities for state transitions. The likelihood of transitioning from the current state $s \in \mathcal{S}$ to the next state $s' \in \mathcal{S}$ is expressed as $P(s'|s, \mathcal{A})$ based on the joint action $\mathcal{A}$. To adapt the MOP to the MARL framework where MUs act as agents, the key POMDP elements of the proposed model are described as follows:

Observation: At time interval t , the observation of each agent k include the downlink rate R_k^D , the uplink bit error probability E_k^U , the terminal energy consumption W_k , the interference noise power $\mathcal{N}_k$, and is expressed as $s_k^t = (R_k^D, E_k^U, W_k, \mathcal{N}_k)$, $k \in \mathcal{K}$.

State: The environment state at time slot t is expressed as $s^t = (s_1^t, s_2^t, \dots, s_K^t)$.

Action: At each time interval t , agents decide on its uplink power allocation, downlink power demand and its total rendering capacity activated. Hence, the action of each agent encompasses the uplink power P_k^U , the downlink power demand P_k^D and the total rendering capacity C_k^R , expressed as $a_k^t = (P_k^U, P_k^D, C_k^R)$, $k \in \mathcal{K}$. At time interval t , the collective action set of all agents is identified as $a^t = (a_1^t, a_2^t, \dots, a_K^t)$.

Reward: A reward function is normally formulated to guide the MARL algorithm towards accomplishing a particular objective. In this paper, the objective of the decision is to achieve the most balanced allocation of downlink power and activating rendering capacity for MUs with their current preferences for MI and terminal energy consumption, while meeting the delay requirement. Thus, the reward for each agent can be expressed as

$$r_k^t = \begin{cases} \omega^1 M_k + \omega^2 W_k, & \frac{C_k^R}{R_k^D} < D_{\max}; \\ 0, & \frac{C_k^R}{R_k^D} \geq D_{\max}. \end{cases} \quad (10)$$

B. M2ERL-UOA Algorithm

Traditional fixed objective weights failed to adapt to different trade-offs between performance objectives quickly, the proposed multi-objective multi-agent evolutionary framework generates the Pareto front policy set about the two goals. This paper denotes the evolutionary learning task by tuple $\mathcal{T} = \langle \mu^\theta, \omega \rangle$, where $\mathcal{T}_i = \{(\mu_i^{\theta_i}, \omega_i)\}_{i=1}^L$, $\omega_i = [\omega_i^1, \omega_i^2]$ and L signifies the amount of the learning tasks for objective weight pairs commencing from the warm-up stage, as well as the number of elite offspring selected during the evolutionary stage for further training. μ^θ is the target policy generating actions by the distributed deployed MUs. As depicted in Fig. 2, leveraging the Multi-Agent Proximal Policy Optimization (MAPPO) framework, the actor network updates the target policy through the multi-objective advantage function collecting sets of trajectories containing state, action, reward and next state.

1) Warm-Up Stage: The warm-up stage (Steps 1–5 in Algorithm 1) generates the MUs' policies forming the initial generation of the population $\mathcal{P}$, where L policies are initialized randomly with L evenly distributed non-negative weight sets of (10), $\omega = \{\omega_1, \dots, \omega_L\}$. The warm-up learning tasks share the same trajectory with the state, action and reward, whereas agents observe unique observations under different perceptions of the dynamic environment. Correspondingly, the actor policy networks are initialized represented by $\mu = \{\mu_1^{\theta_1}, \dots, \mu_L^{\theta_L}\}$. Interacting with the multi-agent environment with the weight sets in each corresponding task for m_w iterations, $L \times m_w$ tasks are trained under the MAPPO framework. Herein, the multi-objective reward and advantage functions are derived during training, facilitating updates to the actor policy and critic value network's parameters. Consequently, it enables us to establish

Algorithm 1: Multi-Objective Multi-Agent Evolutionary Reinforcement Learning Algorithm With User-Object-Attention.

Input: The number of learning tasks and elite offspring L , warm-up iterations m_w , task iterations m_t , task step m_s , generations m_g , the MUs decision intervals $\mathcal{D}$.

- 1: Initialize population $\mathcal{P}$, external Pareto archive EP , and MARL history record $\mathcal{R}$.
- 2: Generate task set $\mathcal{T} = \{(\mu_i, \omega_i)\}_{i=1}^L$ by random initial policies and evenly distributed weight vectors.
- 3: $\mathcal{P}' \leftarrow \text{MAPPO}(\mathcal{T}, m_w, m_s, \mathcal{R})$
- 4: Update $\mathcal{P}$ and EP with $\mathcal{P}'$.
- 5: **for** $\mathcal{G} = 1, 2, \dots, m_g$ **do**
- 6: Fit improvement prediction models $\{\Delta^i\}$ for each policy in $\mathcal{P}$ from data in $\mathcal{R}$.
- 7: $\mathcal{T} \leftarrow \text{TaskSelection}(L, \mathcal{P}, \{\Delta^i\}, EP)$
- 8: $\mathcal{P}' \leftarrow \text{MAPPO}(\mathcal{T}, m_t, m_s, \mathcal{R})$
- 9: Update $\mathcal{P}$ and EP with $\mathcal{P}'$.
- 10: **end for**
- 11: **while** the MUs perform virtual experience and at every decision interval $\mathcal{D}$ **do**
- 12: Get the user preference and calculate the Pareto front point that best fits the current preference.
- 13: The MUs use the policy network contained by the best point and UOA to make decisions.
- 14: **end while**

the offspring population $\mathcal{P}'$ located in the area bounded by the external Pareto archive (EP^1) within the multi-objective space.

2) *Evolutionary Stage:* Advancing into the evolutionary stage (Steps 6–12 in Algorithm 1), we employ the record $\mathcal{R}$ to preserve $\mathcal{P}$ and $\mathcal{P}'$ to find EP comprised of non-dominated policies throughout the evolutionary process, indicating the policy cannot be improved upon in all objectives simultaneously and induce the objective weight pair. For policy evolution and Pareto set approximation, we introduce the predicting policy improvements model, forecasting the goal enhancement of a policy μ with the weight ω after multiple iterations of RL. In this paper, we employ a four-parameter monotonic hyperbolic model to fit the predictions [41], and the model formula is as follows:

$$\Delta_j^i(\omega_j) = U \cdot \frac{e^{x(\omega_j - y)} - 1}{e^{x(\omega_j - y)} + 1} + z, \quad (11)$$

where $\{U, x, y, z\}$ are the four parameters that need to be ascertained. Determining these parameters involves capturing a triplet $(F(\mu), F(\mu'), \omega)$, where $F(\mu_i)$ and $F(\mu'_i)$ signify the policy objectives before and after optimization via multiple iterations of RL. The optimizing weight, denoted as ω , is also included in this process.

Within the iterations involving elite offspring generation, our algorithm identifies and prioritizes the most vital tasks that significantly boost Pareto optimality by assessing hypervolume and

sparsity, which reflect the quality of the Pareto front. Specifically, the algorithm's mission is to pick L learning tasks, indicating the elite offspring, denoted as $\mathcal{T}$. Every task encompasses a pairing of a policy μ with an associated optimization multi-objective weight ω . These tasks within $\mathcal{T}$ undergo processing through multiple iterations of MARL. The evolutionary objective of task selection is to maximize a weighted composite metric $\mathcal{H}(PF) + \varsigma \mathcal{S}(PF)$ [42], where PF represents the Pareto front, $\mathcal{H}(PF)$ denotes the hypervolume of the Pareto front, $\mathcal{S}(PF)$ represents the sparsity of the Pareto front, and ς is a constant less than 0 that minimizes sparsity. We express this optimization problem as:

$$\max_{\mathcal{T}=\{(\mu_i, \omega_i)\}_{i=1}^L} \mathcal{O}(EP, \mathcal{T}) = \mathcal{H}(PF) + \varsigma \mathcal{S}(PF) \quad (12)$$

where $PF = F(EP^*) = \text{Pareto}(F(EP) \cup \{F(\mu_i) + \Delta^i(\omega_i)\})$, EP is the current Pareto archive, and *Pareto* is employed to calculate the Pareto front from a given set of objectives.

The task selection strategy is a knapsack problem that can be solved using dynamic programming. However, in pursuit of a universally applicable solution, we adopt a greedy algorithm, enabling us to navigate the task selection process efficiently. The greedy algorithm maintains a virtual policy set denoted as EP^* , representing the predicted Pareto archive. It consistently selects tasks during iterations that most effectively enhance the Pareto metrics of EP^* . Subsequently, it updates EP^* by merging the predicted offspring policies of the selected tasks. Consequently, in every iteration, those tasks with the potential to markedly enhance the selected Pareto front. The time complexity of the algorithm is related to the number of the population $\mathcal{P}$ and the selected tasks L , i.e., $O(\mathcal{P}L)$. Through this selection process, the chosen tasks undergo optimization using the MAPPO-based multi-objective multi-agent policy gradient algorithm, running in parallel for a specified number of iterations. The outcome of this optimization produces new offspring policies, which update the overall policy population.

Through such repeated iterative calculations, a Pareto front can be obtained. Each point on the Pareto front corresponds to a policy network that can adapt to different system preferences. Thus, in the online stage, we identify the Pareto front point that best aligns with the current system preference. Subsequently, we utilize the corresponding policy network associated with this point to make decisions, thereby maximizing the overall system multi-objective reward.

3) *Multi-Objective Multi-Agent Policy Gradient:* In scenarios where power and rendering capacity need to be determined among multiple MUs, the solution based on MAPPO is formulated and outlined in Algorithm 2. The MAPPO framework is a MARL-based algorithm that involves resource competition between K agents, forming a cooperative game aiming at maximizing overall objectives under use cases such as factory work or virtual schools, where agents interact towards the system gain.

The MAPPO algorithm adopts an actor-critic (AC) framework consisting of the actor network and the critic network. To prevent the policy update from being too large when optimizing the agent objective function, the agent objective function is clipped. The clip function represents the three terms in parentheses.

¹ The EP [41] is employed to store non-dominated policies found during warm-up and evolutionary stage.

Introducing a hyperparameter ϵ , the ratio of the new policy to the old policy can be clipped in the range of $[1 - \epsilon, 1 + \epsilon]$, limiting the probability to a reasonable range. Thus, the actor's objective function is set as follows:

$$J_{\theta}^{\text{clip}} = \mathbb{E}_{\mu^{\theta}} \min \left[\text{clip} \left(\frac{\mu^{\theta}(a^t|s^t)}{\mu^{\theta_{\text{old}}}(a^t|s^t)}, 1 - \epsilon, 1 + \epsilon \right) \tilde{A}^t, \frac{\mu^{\theta}(a^t|s^t)}{\mu^{\theta_{\text{old}}}(a^t|s^t)} \tilde{A}^t \right] + \psi S(s^t), \quad (13)$$

where $\min$ indicates to take the smaller formula between the first item and the second item. $S(s^t)$ represents the policy entropy of the state, adding exploratory during the training phase can help avoid convergence to a local optimum. The actor network parameters θ encapsulated $\{(\theta_{i1}, \dots, \theta_{ik}, \dots, \theta_{iK})\}_{i=1}^L$ for all agents. ψ denotes a hyperparameter that balances exploration and exploitation. $\tilde{A}^t$ is usually estimated in the form of generalized advantage estimation (GAE) expressed by

$$\tilde{A}^t(s^t, a^t) = \sum_{l=0}^{\infty} (\gamma \kappa)^l \sigma_V^{t+l}, \quad (14)$$

$$\sigma_V^t = r^t + \gamma V_{\phi}(s^{t+1}) - V_{\phi}(s^t), \quad (15)$$

where V_{ϕ} is the value function of the critic network and the bias and variance of the estimate are balanced in a weighted manner by introducing the parameter κ .

Throughout the phase of centralized training, agents observe each other's states and actions. The central coordinator, represented by the critic network, undertakes the computation of the centralized action-value function $Q(s^t, a_1^t, a_2^t, \dots, a_K^t | \phi)$ utilizing comprehensive global state information encompassing actions and observations of all agents. This centralized Q function facilitates the evaluation of actor actions from a holistic viewpoint, thereby enhancing decision-making regarding actions. Following this, updates to the parameters ϕ of the critic network occur through the minimization of loss expressed as follows:

$$V_{\phi} = \mathbb{E} [Q(s^t, a_1^t, a_2^t, \dots, a_K^t | \phi) - y^{\text{MAPPO}}]^2, \quad (16)$$

where

$$y^{\text{MAPPO}} = r^t + \gamma Q'(s^{t+1}, a_1^{t+1}, a_2^{t+1}, \dots, a_K^{t+1} | \phi') \quad (17)$$

in which $s^t = (s_1^t, s_2^t, \dots, s_K^t)$ and ϕ represent the critic network parameter consisting of $\{(\phi_{i1}, \dots, \phi_{ik}, \dots, \phi_{iK})\}_{i=1}^L$ for all agents. The refreshed states for the target network are denoted as s^{t+1} , while ϕ' signifies the updated parameters of the evaluation network.

In the execution phase, the critic network remains inactive, enabling the well-trained actor network combined with the distributed UOA mechanism to engage in decision-making, relying on local MU observations. This implies an exclusive forward propagation process utilizing ubiquitous distributed computing capabilities, substantially cutting down the time and computational resources expended in contrast to the training period. Each MU's actor agent, equipped with fine-tuned parameters, acquires a nearly globally optimal action without knowledge of

Algorithm 2: MAPPO-Based Policy Gradient.

Input: Task set $\mathcal{T}$, episode m_e , step m_s , MARL history record $\mathcal{R}$.
Output: Offspring population $\mathcal{P}'$.
1: **for** each task $(\mu_i^{\theta_i}, \omega_i) \in \mathcal{T}$ **do**
2: **for** episode = 1, 2, ..., m_e **do**
3: Initialize a memory buffer $D = \emptyset$ and the state of all MUs.
4: **for** step = 1, 2, ..., m_s **do**
5: **for** all agents k **do**
6: $a_k^t = \mu_{ik}^{\theta_{ik}}(s_k^t; \theta_{ik})$.
7: $(C_{k,1}^R, \dots, C_{k,N_k}^R) \leftarrow \text{UOA}(T_{k,n}^R, C_k^R)$
8: $v_k^t = V_{ik}^{\phi_{ik}}(s_k^t, a_k^t; \phi_{ik})$.
9: **end for**
10: Get reward r^t and next state s^{t+1} according to resource allocation result.
11: Store $[s^t, s_k^t, a^t, r_k^t, s^{t+1}, s_k^{t+1}]$ in D .
12: **end for**
13: Compute advantage estimate $\tilde{A}^t$ via GAE.
14: Split trajectory into chunks of length B
15: **for** all agents k **do**
16: **for** mini-batch $b = 1, \dots, B$ **do**
17: $batch \leftarrow$ random mini-batch from D with all agent data.
18: Adam update θ_{ik} with data $batch$ by (13).
19: Adam update ϕ_{ik} with data $batch$ by (16).
20: **end for**
21: **end for**
22: Collect the result policy $\mu_i^{\theta_i'}$ in $\mathcal{P}'$.
23: Store $(F(\mu_i^{\theta_i}), F(\mu_i^{\theta_i'}), \omega_i)$ in $\mathcal{R}$.
24: **end for**
25: **end for**

the information held by other agents, guaranteeing the MUs' privacy.

In this paper, the points that establish the Pareto front contain a non-dominated solution and a preference for the two indicators and the multi-agent policy network. MUs can input their own state into the policy network and obtain the output policy. The policies of all MUs in the system are combined to create a joint policy. We can prove that the joint policy on the Pareto front is the Nash equilibrium [43], [44], [45], [46], [47].

Theorem 1: For each point on the Pareto front, a process of optimization to a non-dominated solution can be obtained, and the weights of the two indicators are fixed in this process.

Proof: Please refer to Appendix A, available online. $\square$

Theorem 2: In the case that the weights of the two indicators are fixed, the policy network trained to convergence generates the optimal policy $\mu_k^*(s^t)$, representing a Nash equilibrium point in the K-agent stage game.

Proof: Please refer to Appendix B, available online. $\square$

According to Theorem 1, for any point on the Pareto front, an iterative chain $\{(\mu_0^{\theta_0}, \omega_0), \dots, (\mu_L^{\theta_L}, \omega_L)\}$ (see Appendix A for more detail, available online) can be found. Also, the weight of MI and terminal energy consumption of each node

on the iterative chain is fixed. For an iterative chain, the model parameters corresponding to the start node are trained step by step and finally converge to the model parameters corresponding to the end node. In particular, the policy network on the Pareto front is convergent. According to Theorem 2, a joint policy from the policy network trained to convergent is the Nash equilibrium. We thus prove that the joint policy on the Pareto front is the Nash equilibrium.

4) *UOA-Based Optimal Rendering Capacity Allocation*: Importing the user-object-attention level (UOAL) dataset [9], [48] enables the acquisition of the quantitative attention value of each MU to each object. The more interested the MU is in an object, the higher the attention level of that object will be. Rendering capacity should be allocated to objects with attention to guarantee the MI. Thus, we subsequently formulate an optimal rendering capacity allocation scheme based on the user-object-attention mechanism, addressing the maximization problem as follows:

$$\begin{aligned} \max_{C_{k,1}^R, \dots, C_{k,N_k^O}^R} \quad & \sum_{n=1}^{N_k^O} T_{k,n} \ln \left(\frac{C_{k,n}^R}{C_{th}^R} \right) \\ \text{s.t.} \quad & \begin{cases} C_{k,n}^R > C_{th}^R, \forall n, \\ \sum_{n=1}^{N_k^O} C_{k,n}^R \leq C_k^R, \end{cases} \end{aligned} \quad (18)$$

where C_k^R represents the total rendering capacity allocated to the k th MU. The constraint condition is that the sum of the rendering capacity allocated to each virtual object of the k th MU cannot be greater than C_k^R , and the rendering capacity allocated to each virtual object must be greater than the minimum threshold C_{th}^R . This paper uses the KKT algorithm to allocate rendering capacity.

Theorem 3: In pursuit of maximizing the MI, the allocation of rendering capacity for each object is determined as follows:

$$C_{k,n}^R = \max \left\{ T_{k,n} \frac{1}{\xi^*}, C_{th}^R \right\}, \quad (19)$$

where ξ^* is acquired by solving the following function

$$\sum_{n=1}^{N_k^O} \max \left\{ T_{k,n} \frac{1}{\xi^*}, C_{th}^R \right\} = C_k^R. \quad (20)$$

Proof: Please refer to Appendix C, available online. $\square$

Utilizing Theorem 3 as a foundation, we propose an optimal rendering capacity allocation algorithm as shown in Algorithm 3. This algorithm demonstrates effective convergence and yields the optimal rendering capacity allocation scheme. Its time complexity is linearly related to the number of virtual objects, which is $O(N_k^O)$.

V. PERFORMANCE EVALUATION

A. Simulation Settings

The simulation environment consists of a BS and MUs. The total downlink power that the BS can allocate is 240 dBm, and the maximum uplink power and total rendering capacity that can be allocated to each MU are 30 dBm and 500 K, respectively. In this context, we establish that a 1 K resolution corresponds to a

Algorithm 3: Rendering Capacity Allocation With UOA.

Input: The user-object-attention values $T_{k,n}$, the total rendering capacity C_k^R .

Output: The optimal rendering capacity allocation scheme, $C_{k,1}^R, \dots, C_{k,N_k^O}^R$.

- 1: Initialize two temporary variables $T_{td} = \sum_{n=1}^{N_k^O} T_{k,n}$ and $j = 0$.
- 2: Start by assuming $C_{k,n}^{R*} = T_{k,n} \frac{1}{\xi^*}$ for all n .
- 3: Calculate $\xi^* \leftarrow \sum_{n=1}^{N_k^O} T_{k,n} / C_k^R$, and then $C_{k,n}^{R*} = T_{k,n} / \xi^*$ for every n .
- 4: **while** the minimum of $C_{k,n}^{R*} < C_{th}^R$ **do**
- 5: Assign rendering capacity C_{th}^R to the object with the minimum $C_{k,n}^{R*}$.
- 6: $j \leftarrow j + 1$.
- 7: $T_{td} \leftarrow T_{td} - T_{\min}$. $T_{\min}$ represents the attention value of the object with the minimum rendering power.
- 8: Re-calculate $\xi^* \leftarrow T_{td} / (C_k^R - j * C_{th}^R)$.
- 9: Re-calculate $C_{k,n}^{R*} \leftarrow T_{k,n} / \xi^*$ for each object with undetermined rendering capacity.
- 10: **end while**
- 11: **return** The optimal rendering capacity allocation scheme, $C_{k,1}^R, \dots, C_{k,N_k^O}^R$.

TABLE III
SIMULATION PARAMETERS

System Parameters		Training Parameters [41]	
Parameters	Value	Parameters	Value
$T_{k,n}$ [49]	[0, 5]	L	6
C_{th}^R [5]	12 K	m_w	80
C_{max}^R [9]	500 K	m_t	20
B [50]	500 MHz	ς	-1
R_{max}^D [51], [5]	1 Gbps	Iteration step m_s	2048
E_{max}^U [52]	0.0001	Learning rate	$3e - 4$
$\mathcal{N}$ [9, 53]	[3 dBm, 6 dBm]	Discount factor γ	0.995
P_{total}^D	240 dBm	gae lambda	0.95
P_{max}^U	30 dBm	num minibatch	32
D_{max} [5]	58 ms	entropy coefficient	0
$\mathcal{V}$ [5]	12 bits/pixel	value loss coefficient	0.5

pixel resolution of 960×480 . To achieve an optimal experience, it is recommended to have a resolution surpassing 12 K [5]. Moreover, the attention $T_{k,n}$ of the MU k to the virtual object n is an integer and satisfies $0 \leq T_{k,n} \leq 5$. The interference noise power of the environment changes dynamically between 3 dBm and 6 dBm. Particularly, the settings of the simulation parameters are listed in Table III.

We compare the performance gain of the M2ERL-UOA algorithm, with the three widely used evolutionary algorithms as the benchmarks. To demonstrate the effectiveness of dimensionality reduction, this paper also compares dimensionality reduction algorithms with a non-dimensionality reduction algorithm called M2ERL.

- DCNSGA-III: DCNSGA-III [54] is an algorithm for multi-objective optimization that transforms the problem into a

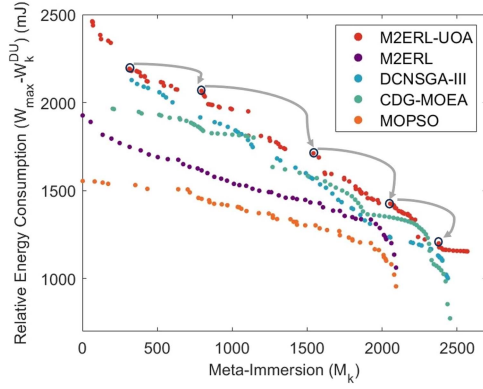


Fig. 3. Pareto front comparison of the four algorithms.

dynamic constrained multi-objective optimization problem to handle both objectives and constraints simultaneously. It uses ϵ -feasible solutions and an adapted NSGA-III [55] framework to maintain diversity and navigate infeasible regions, showing strong performance.

- PFG-MOEA: PFG-MOEA [56] is a multi-objective evolutionary algorithm that uses a Pareto Front Grid (PFG) to guide the search. It employs a statistical approach for nadir point estimation to reduce computation time and introduces a grid-based knee point selection method for environmental selection. This results in a grid-based decomposition multi-objective evolutionary algorithm.
- MOPSO: MOPSO [57] is a multi-objective particle swarm optimization algorithm. It is based on a particle swarm optimization algorithm and continuously optimizes the objective function value by iteratively updating the particle swarm in the search space.

Adopting the above baseline algorithms to solve the proposed multi-agent problem, we take the set of actions that all agents decide parameters for the baseline algorithm to improve. In this paper, we ensure uniformity by assigning identical values to all shared hyperparameters. Furthermore, we execute all algorithms with an equal number of simulation steps to facilitate a fair comparison between the baseline algorithm and our proposed approach.

B. Pareto Front and System Reward Comparison

In Fig. 3, a visual representation of the Pareto fronts generated by five distinct algorithms is depicted. In order to better observe the quality of the Pareto front, this paper makes a positive transformation of the considered indicators. The energy consumption is converted to relative energy consumption, which is obtained by subtracting the actual energy consumption from the maximum energy consumption of the terminal. It is evident from Fig. 3 that the three algorithms, M2ERL-UOA, PFG-MOEA, and DCNSGA-III, are relatively close in terms of Pareto front quality. Notably, our proposed algorithm stands out by generating a high-quality Pareto front. This demonstrates that our algorithm can identify regions where the Pareto front is near-optimal, as well as areas that can be further enhanced

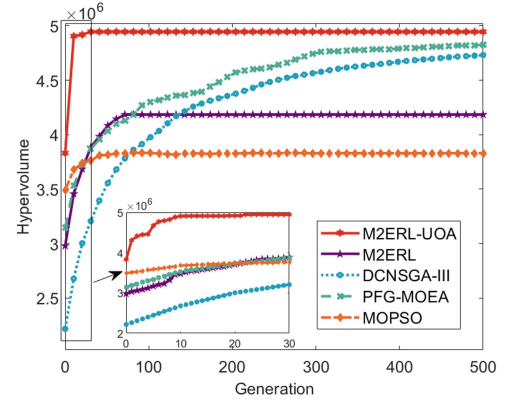


Fig. 4. Hypervolume change of Pareto front during training of different algorithms.

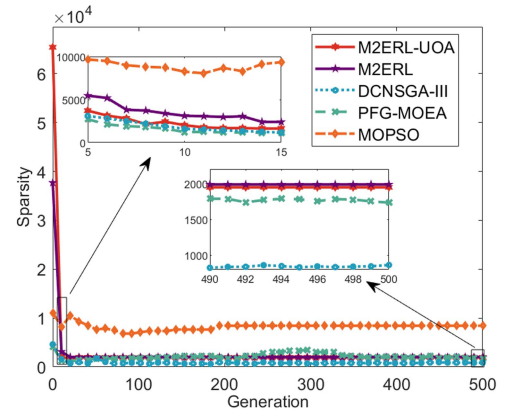


Fig. 5. Sparsity change of Pareto front during training of different algorithms.

through policy improvement of the predictive model. As a result, optimal tasks can be strategically selected to create a high-quality Pareto front and achieve better trade-offs between different objectives. In contrast, the M2ERL algorithm struggles to obtain a high-quality Pareto front due to the large action space.

To validate the necessity of dimensionality reduction, we compared the Pareto front quality of the M2ERL-UOA algorithm with and without dimensionality reduction and the other three baseline algorithms in the training process, as shown in Figs. 4 and 5. This paper selects two indicators, hypervolume and sparsity, to reflect the quality of the Pareto front. It can be seen from both metrics that using the dimensionality reduction technique can greatly improve the approximation of the Pareto front. Benefiting from the technique that only needs to determine the total rendering capacity allocated to each user, and the usage of iterative optimization to allocate the total rendering capacity to each virtual object, the action dimension can be greatly reduced, enabling the acquisition of a high-quality Pareto front. Noted that compared to the evolutionary algorithms, M2ERL-UOA and M2ERL converge with fewer iterations. This improved convergence speed can be attributed to the utilization of RL and prediction models in the proposed algorithms to generate high-quality offspring. As a result, the convergence

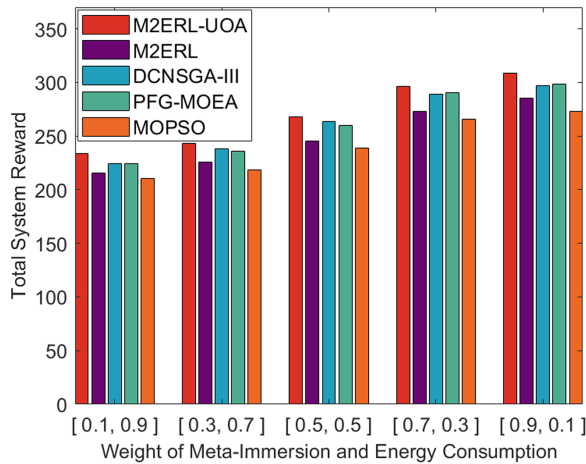


Fig. 6. Total system reward of different algorithms under different user preferences.

process is accelerated, leading to more efficient and effective optimization outcomes. From the perspective of sparsity, there is no significant difference between reduced and non-reduced dimensions. This shows that the algorithm proposed in this paper is more universal and it can ensure a certain continuity of the Pareto front when the action dimension is different. Compared to the other three baseline algorithms, the initial sparsity of the M2ERL-UOA algorithm is relatively high. However, after subsequent convergence, the algorithm achieves a significantly reduced sparsity, providing evidence of the effectiveness of the proposed algorithm. In addition, our algorithm can also show the advantages described above when trained on multiple different environmental parameters.

Fig. 6 displays the maximum total system reward after the convergence of different algorithms under various user preferences with fixed interference noise power. Fig. 3 illustrates the policy network switching process of the proposed algorithm with different user preferences. The data presented in these figures demonstrate that the proposed algorithm exhibits a remarkable ability to flexibly adjust to the preferences of users and attain greater system rewards than other algorithms. This superiority can be attributed to the fact that the proposed evolutionary learning algorithm effectively identifies higher-quality Pareto fronts by leveraging the extension of MARL-based algorithms to guide the learning process.

To validate the superiority of the proposed algorithm in dynamic environments, we considered a scenario where user preferences and interference noise power change every 15 time intervals, simulating 30 intervals in total. Fig. 7(a) shows the variation in total system reward over time for five algorithms under this scenario. It is evident from the figure that the other three evolutionary algorithms lack the adaptability of RL and cannot retain strategies for different preferences and interference noise power levels. When the environment changes, these algorithms must restart from scratch, meaning they are still in the training phase to adapt to the environment, resulting in relatively lower system rewards. Although the M2ERL algorithm adapts quickly to changes, its large action space makes it difficult to train a

high-quality Pareto front, leading to lower system rewards. In contrast, our algorithm is flexible, quickly adapting to changes in user preferences and interference noise power. It helps MUs swiftly switch strategies, achieving a more balanced trade-off between MI and terminal energy consumption. Fig. 7(b) and (c) describe the Pareto fronts corresponding to two different noise power levels. When the interference noise power is changed, the proposed algorithm achieves the highest quality Pareto front. Despite observing an increasing trend in system rewards for other evolutionary algorithms over time, they still fail to surpass the overall system reward achieved by the proposed algorithm upon convergence. This comparison highlights the superior performance of the proposed algorithm in producing a top-quality Pareto front, showcasing its effectiveness in dealing with dynamic environment.

C. Matrix Comparison

Fig. 8 presents the average decision time and standard deviation of decision time for different algorithms. It is evident from the figure that the M2ERL-UOA and M2ERL algorithms perform remarkably well in terms of average decision time, while the DCNSGA-III and PFG-MOEA algorithms have moderate average decision times, and the MOPSO algorithm has the longest. Moreover, the decision time standard deviation of the M2ERL-UOA and M2ERL algorithms is smaller than the other three, indicating more stable decision times. This difference arises as the other three evolutionary algorithms, by their innate nature, fail to preserve situational strategies amidst dynamically shifting environments. This paper considers conducting only a limited number of training iterations during each decision-making instance and continuing training in subsequent decisions, thus distributing the algorithm's overall training time across each decision to enhance the user experience. Environmental alterations necessitate these baseline algorithms to recommence training to conform to the altered circumstances, whereas our proposed algorithm merely entails the expedient selection of the optimally fitting policy network from the Pareto front for immediate decision execution, devoid of supplementary training periods, which drastically curtails decision latency. Compared to M2ERL, the M2ERL-UOA algorithm has a smaller action space and fewer neural network layers, resulting in faster average decision times and a more user-friendly experience.

The cumulative distribution function (CDF) comparison of delay for the five algorithms is shown in Fig. 9. The maximum delay threshold is set to 58 ms, as referenced in [58]. From the figure, it is evident that in a dynamic environment, the delay for the other four baseline algorithms exceeds the threshold, whereas only the proposed algorithm fully meets the delay requirements. This indicates that the other algorithms fail to effectively allocate power and rendering capacity, often resulting in either insufficient power or excessive rendering capacity being assigned to users, leading to delays that exceed the threshold and severely impact user experience. The three evolutionary algorithms experience excessive delays due to the need for retraining when interference noise and user preferences change.

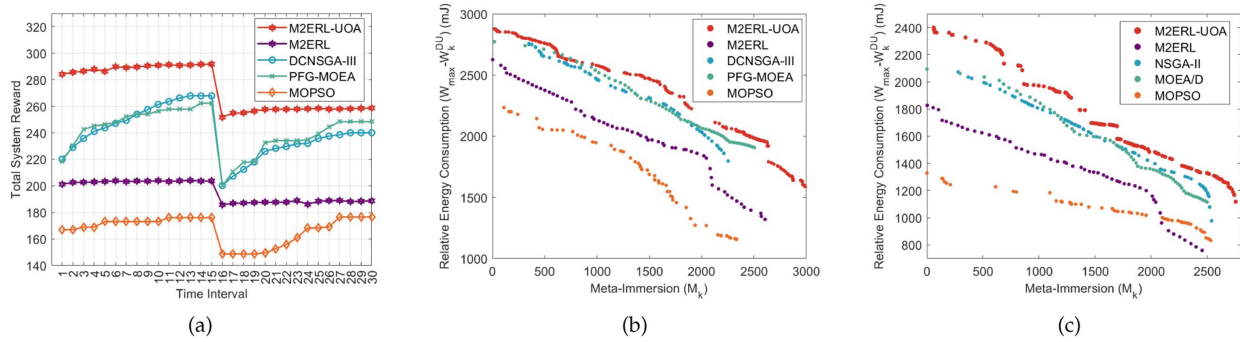


Fig. 7. Online performance comparison of five algorithms. (a) Total system reward. (b) The Pareto front in the first 15 time interval environment. (c) The Pareto front in the post 15 time interval environment.

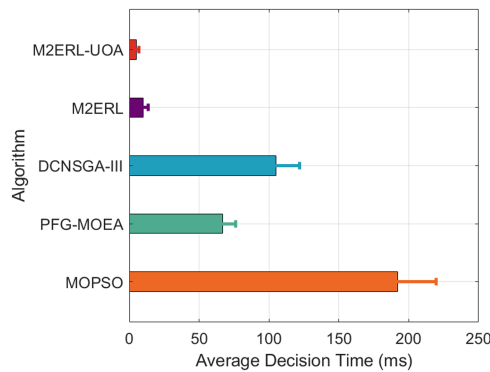


Fig. 8. Average decision time under different algorithms.

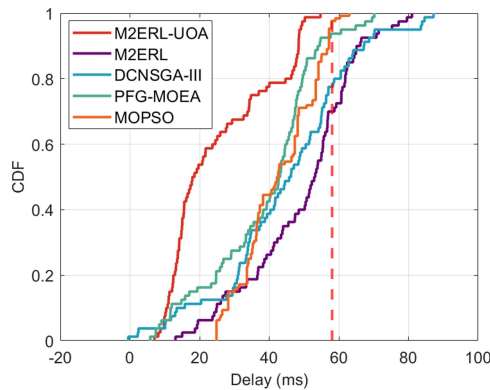


Fig. 9. Cumulative probability distribution diagram of delay in the dynamic environment.

The M2ERL algorithm produces a low-quality Pareto front due to its large action space, causing delays to exceed the threshold. The M2ERL-UOA algorithm empowers MUs to reach optimal decisions promptly, effectively minimizing delay and improving the realism of their experience. Consequently, our proposed algorithm can flexibly adapt to dynamic environments, providing MUs with a consistently high-quality virtual experience.

Fig. 10 shows the energy consumption over time when the MI is no less than 200. It can be noted that the proposed algorithm

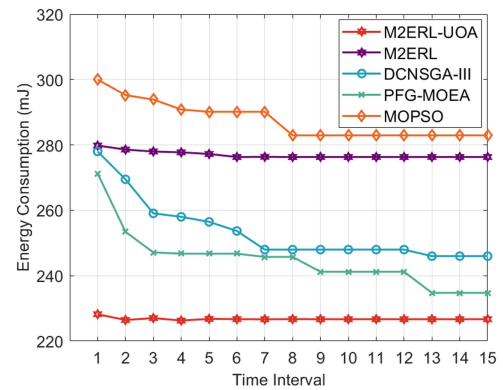


Fig. 10. Energy consumption under the condition that the MI requirements are met.

provides MUs with a strategy that minimizes energy consumption while satisfying the MI requirement, whereas the other three evolutionary algorithms require a longer training time. The M2ERL algorithm produces a lower-quality Pareto front and fails to obtain an optimal energy consumption strategy under the MI constraint. The advantage of the proposed algorithm is also demonstrated in this scenario, as it calculates a high-quality Pareto front, offering decision-makers more options to balance different objectives.

Fig. 11 shows the MI of MUs under three different resource allocation algorithms with the same total rendering capability. We compare three rendering capacity allocation algorithms: the proposed optimal algorithm, a fairness allocation algorithm, and an unconsidered allocation algorithm. The fairness allocation algorithm assigns equal rendering capacity to each virtual object. It ensures an equitable distribution of rendering capacity, irrespective of the MUs' attention level to virtual objects. On the other hand, the unconsidered allocation algorithm randomly assigns rendering capacity to virtual objects. The results depicted in the figure illustrate that the proposed optimal rendering capacity allocation algorithm enhances MI. In contrast, the fairness algorithm allocates the same rendering capacity to all virtual objects, leading to inefficient resource utilization by rendering objects that the MU may not be interested in. We introduce an innovative approach by decomposing the problem model,

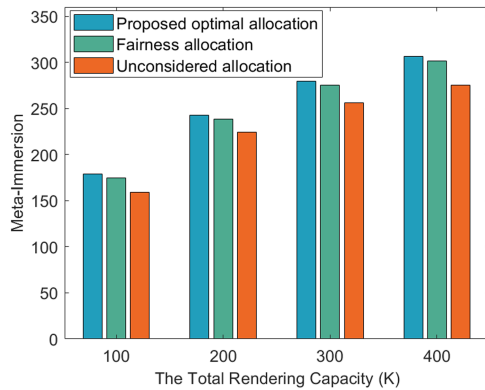


Fig. 11. Comparison of the total rendering capacity allocated by different algorithms.

utilizing RL to determine the MU's overall rendering capability, and employing the KKT algorithm for further object rendering capacity allocation. This not only reduces the dimensionality of the original problem but also enhances resource utilization efficiency, ultimately providing an improved MI experience for MUs.

VI. CONCLUSION

In this paper, we propose the M2ERL-UOA algorithm to solve the problem of how the base station allocates system resources to multiple MUs to achieve the best balance in the case of dynamic changes in interference noise power and system preference. Our algorithm has two stages. In the offline stage, we utilize a prediction-guided evolutionary learning algorithm to discover a high-quality collection of policies within the Pareto set. Furthermore, we use the KKT algorithm to allocate rendering capacity to virtual objects of different attention and prove that the policies on the Pareto front are Nash equilibrium. In the online stage, the user will select the most appropriate policy network on the Pareto front according to the current preference to make decisions. Finally, we verify that the algorithm can flexibly allocate system resources to multiple MUs in the dynamic environment through numerical simulations. Overall, this work provides a solution to the multi-objective multi-agent optimization problem in the Metaverse. In the future, our vision entails the introduction of meta-learning to ensure the ability to quickly and precisely accommodate evolving user preferences.

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